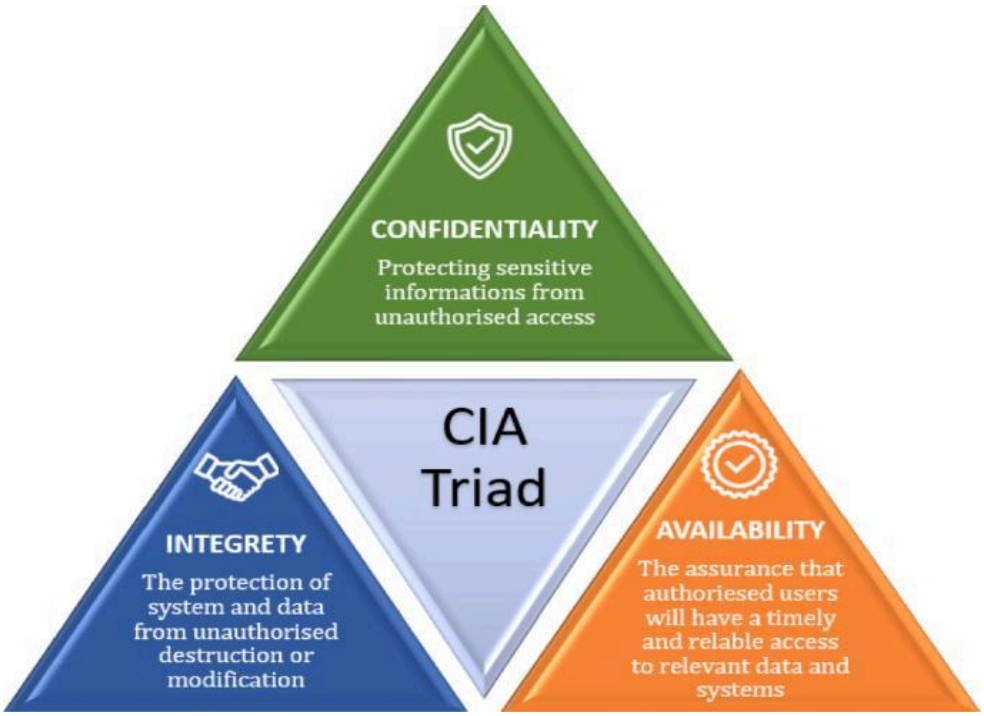
Slip 01

Q. 1 Explain the CIA triad (Confidentiality, Integrity, Availability) with suitable examples

CIA Triad (Confidentiality, Integrity, Availability)

The CIA Triad is a fundamental model of cybersecurity that helps protect information and computer systems. It consists of three main principles: Confidentiality, Integrity, and Availability.



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1. Confidentiality

Meaning: Confidentiality ensures that information is accessible only to authorized users.

- It protects sensitive information from unauthorized access.
- Techniques include passwords, encryption, and access control.

Example: A student's examination marks can be viewed only by authorized teachers and college administrators, not by other students.

2. Integrity

Meaning: Integrity ensures that information remains accurate, complete, and unchanged by unauthorized persons.

- It prevents unauthorized modification or deletion of data.
- Techniques include hashing, digital signatures, and access control.

Example: If a student's marks are 85, an unauthorized person must not be able to change them to 95.

3. Availability

Meaning: Availability ensures that information and computer systems are accessible to authorized users whenever needed.

- It prevents disruptions to services and systems.
- Techniques include regular backups, system maintenance, and protection against DoS attacks.

Example: A college website should remain accessible to students when they check examination results, even when many students access it simultaneously.

Summary Table

Principle	Main purpose	Example
Confidentiality	Protect data from unauthorized access	Password-protected student records
Integrity	Maintain data accuracy	Preventing unauthorized changes to marks
Availability	Ensure access when required	Keeping the college website operational

Conclusion: The CIA Triad forms the foundation of information security. Confidentiality protects data privacy, Integrity maintains data accuracy, and Availability ensures access to information when needed. All three principles are essential for securing computer systems and information.

Q2)
B
Write a Python program to simulate a phishing login page that collects username and password and stores them in a file.

```
# Phishing Login Simulation
# Educational Purpose Only

print("====================================")
print("     PHISHING SIMULATION DEMO")
print("       EDUCATIONAL PURPOSE")
print("====================================")

username = input("Enter Username: ")
password = input("Enter Password: ")

# Store the entered dummy data in a local file
with open("simulation_data.txt", "a") as file:
    file.write("Username: " + username + "\n")
    file.write("Password: " + password + "\n")
    file.write("----------------------------\n")

print("\nLogin processing...")
print("This is a simulated phishing page.")
print("Demo data has been stored in simulation_data.txt")
```

OR

Q.2 Write a Python program to implement a dictionary attack using a list of common passwords to check whether a target password exists in the list.

```
# Dictionary Attack Simulation

def dictionary_attack(target):
    passwords = [
        "admin123", "password", "123456", "qwerty",
        "welcome123", "computer@123", "hello123",
        "student123", "india@123", "python123"
```

```
    ]

    attempts = 0

    for pwd in passwords:
        attempts += 1
        print("Attempt", attempts, ":", pwd)

        if pwd == target:
            print("Password matched!")
            print("Total attempts:", attempts)
            return

    print("Password not found in dictionary.")
    print("Total attempts:", attempts)


target = input("Enter target password: ")
dictionary_attack(target)
```